

CHATHAM, MA, USA

WMO: 725069

Lat: 41.688N

Lon: 69.993W

Elev: 68

StdP: 14.66

Time Zone: -5.00 (NAE)

Period: 95-19

WBAN: 94624

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.7	16.4	-3.1	4.7	16.0	1.2	5.9	19.7	25.6	37.4	22.4	37.3	8.3	300	0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	83.4	73.4	80.9	72.0	78.7	70.8	75.6	80.2	74.3	78.3	73.0	76.6	7.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.9	126.7	78.4	72.8	122.1	76.9	71.9	118.1	75.8	39.0	80.5	37.7	78.1	36.6	76.9	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.0	17.6	15.9	DB	7.5	88.6	5.9	3.8	3.3	91.3	-0.2	93.6	-3.5	95.7	-7.8	98.5	(4)
(5)			WB	5.8	77.7	5.5	1.8	1.8	79.0	-1.4	80.1	-4.5	81.1	-8.5	82.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.1	32.0	32.3	37.7	46.0	55.1	64.1	71.2	70.7	64.6	54.9	45.9	37.5		
	DBStd	15.31	8.71	7.63	6.75	5.66	5.73	5.74	4.71	4.56	5.31	6.74	7.34	7.69		
	HDD50	2181	559	496	383	143	16	0	0	0	0	29	162	393		
	HDD65	5581	1023	916	847	569	311	84	7	7	70	320	573	854		
	CDD50	2588	0	0	1	25	174	423	658	643	440	180	40	4		
	CDD65	512	0	0	0	0	4	57	200	185	60	6	0	0		
	CDH74	1996	0	0	0	5	20	186	899	765	115	6	0	0		
	CDH80	314	0	0	0	0	5	31	165	106	7	0	0	0		
(14) Wind	WSAvg	7.5	7.8	8.1	8.7	8.5	7.7	7.1	6.6	6.2	6.7	7.1	7.5	7.7		
(15) Precipitation	PrecAvg	38.9	3.4	3.0	3.7	3.6	2.9	2.9	2.2	2.8	3.3	3.3	3.7	3.9		
	PrecMax	49.8	6.0	6.8	8.8	7.2	6.7	10.4	5.9	8.5	7.5	7.7	8.0	7.0		
	PrecMin	22.7	0.7	1.3	0.4	1.1	0.6	0.3	0.4	0.0	0.9	0.6	0.8	0.5		
	PrecStd	7.6	1.4	1.3	2.1	1.5	1.7	2.0	1.1	1.9	1.9	1.8	1.9	1.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.3	51.9	57.2	67.2	76.2	83.9	88.1	86.1	80.9	73.3	63.6	57.4		
		MCWB	52.1	49.0	49.3	55.4	64.4	71.6	74.5	74.9	72.2	66.0	58.0	54.3		
	2%	DB	49.7	48.4	52.5	60.0	70.1	78.8	84.2	82.6	77.1	69.5	61.1	54.3		
		MCWB	47.8	46.1	47.4	52.4	61.6	68.3	73.7	73.6	69.7	65.1	58.1	52.8		
	5%	DB	47.0	45.8	49.7	56.8	66.3	75.3	81.3	80.7	74.8	66.6	58.7	52.0		
		MCWB	44.5	43.2	45.8	50.4	59.1	66.6	72.0	72.5	68.6	62.9	55.9	50.5		
	10%	DB	44.4	43.3	47.1	54.4	63.6	72.7	78.9	78.4	72.7	64.3	56.6	49.6		
		MCWB	41.8	40.3	43.8	49.5	57.5	65.6	71.1	71.1	67.6	61.0	53.7	47.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.3	50.5	53.1	59.4	66.6	73.4	77.7	77.7	74.6	68.8	61.0	56.0		
		MCDB	52.9	51.0	54.5	63.8	73.5	81.0	85.5	83.7	79.0	71.4	62.1	56.5		
	2%	WB	48.4	46.9	49.1	55.1	63.6	71.0	75.7	75.9	72.3	66.1	58.9	53.3		
		MCDB	49.3	47.9	51.1	58.3	68.0	76.5	80.9	80.2	75.2	68.0	60.6	53.9		
	5%	WB	45.5	44.3	46.6	52.9	61.3	69.3	74.3	74.6	70.7	64.1	56.5	50.8		
		MCDB	46.8	45.6	48.3	54.9	64.8	73.2	78.7	78.3	73.3	65.9	58.4	51.6		
	10%	WB	42.2	41.2	44.4	50.7	59.0	67.4	73.0	73.3	69.2	62.0	54.2	47.8		
		MCDB	44.1	43.2	46.5	53.3	62.5	70.8	76.8	76.6	71.7	64.0	56.0	49.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.9	12.8	11.7	12.0	12.4	12.5	12.8	12.7	12.4	12.4	11.9	12.0		
		MCDBR	13.7	14.1	14.3	15.5	16.7	16.0	15.4	14.2	13.6	13.2	11.9	13.5		
		MCWBR	13.7	13.1	11.3	10.0	10.0	9.0	8.1	8.2	8.9	10.3	10.6	14.0		
	5% WB	MCDBR	13.4	13.8	12.9	13.1	15.3	13.9	13.1	12.5	11.5	12.1	11.1	13.1		
		MCWBR	14.2	14.2	11.8	10.5	9.9	8.7	7.7	7.9	8.8	10.9	11.2	14.4		
(40) Clear-Sky Solar Irradiance	taub		0.314	0.320	0.338	0.375	0.408	0.456	0.481	0.440	0.374	0.350	0.323	0.310		
	taud		2.511	2.496	2.482	2.396	2.349	2.245	2.186	2.297	2.457	2.521	2.561	2.559		
	Ebn at Noon		266	283	289	284	276	262	253	261	273	268	262	259		
	Edh at Noon		23	27	31	36	39	43	45	40	31	26	22	21		
(44) All-Sky Solar Radiation	RadAvg		530	819	1161	1517	1722	1845	1902	1715	1394	945	614	453		
	RadStd		42	75	92	162	167	165	104	90	100	78	47	46		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	+0.97	+0.92	N/A	N/A	N/A	N/A
(47) Regional (49 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+175	+85

Nomenclature: See separate page